

Computability: Exercise 9

Due: June 9th, 2026

Question 1

For any positive integer k , we define the k -COLOURING problem as follows:

$$\{\langle G = (V, E) \rangle : \forall v \in V, C(v) \in [k] \text{ and } \forall e = \{u, v\} \in E, C(u) \neq C(v)\}$$

Where $C : V \rightarrow [k]$ is a mapping between vertices and colours.

Namely, a graph belongs to the language if it is possible to colour all vertices such that no neighbouring vertices have the same colour.

- Show that for every k , k -COLOURING $\leq_p (k+1)$ -COLOURING.
- Show that 2-COLOURING is in P.

Question 2

Prove that $3\text{-SAT}^{\geq 3} = \{\langle \varphi \rangle : \varphi \text{ is a 3-CNF boolean formula with at least 3 satisfying assignments}\}$ is NP-complete.

Question 3

We define k -NAE-SAT (Not All Equal) problem in the following way:

k -NAE-SAT = $\{\langle \varphi \rangle : \varphi \text{ is a } k\text{-CNF formula and there exists a satisfying assignment s.t.}$

for each clause $c_j = (l_1^j \vee l_2^j \vee \dots \vee l_k^j)$ at least one literal value is "false"

In other words, there is an assignment where at least one literal achieves "true", and at least one literal achieves "false" for each clause.

- Show that k -NAE-SAT is in NP using a polynomial verifier.
- Prove that $3\text{-SAT} \leq_p 4\text{-NAE-SAT}$
- Prove that $4\text{-NAE-SAT} \leq_p 3\text{-NAE-SAT}$ and conclude that 3-NAE-SAT is NP-complete.
Hint: Consider the following reduction of a literal - $(l_1 \vee l_2 \vee l_3 \vee l_4) \rightarrow (l_1 \vee l_2 \vee s) \wedge (l_3 \vee l_4 \vee \bar{s})$.

Question 4

Given an undirected graph G , we define the following languages:

SPATH = $\{\langle G, u, v, k \rangle \mid G \text{ contains a simple path of length at most } k \text{ from } u \text{ to } v\}$

LPATH = $\{\langle G, u, v, k \rangle \mid G \text{ contains a simple path of length at least } k \text{ from } u \text{ to } v\}$

Where a path is simple if it does not repeat vertices (every vertex appears at most once), and the size of the length of a path is the number of edges in the path.

- a. Show that $\text{SPATH} \in P$.
- b. Show that LPATH is NP-Complete.

Question 5

The feasibility problem of *Integer Linear Programming* is defined as follows: given a set of m linear constraints:

$$\sum_{j=1}^n a_{i,j} \cdot x_j \leq b_i,$$

where $1 \leq i \leq m$. We have to decide whether there exists a $-1/0/1$ assignment to $x_1, \dots, x_n$ such that all constraints are satisfied. In matrix notation, an instance of the feasibility problem is given by a pair of an $m \times n$ matrix over $\mathbb{Z}$, denoted A , and a column vector of size m over $\mathbb{Z}$, denoted b , and we have to decide if there is some column vector of size n over $\{-1, 0, 1\}$, denoted x , such that $Ax \leq b$.

Let

$$ILP = \{\langle A, b \rangle : A \in \mathbb{Z}_{m \times n}, b \in \mathbb{Z}^m \text{ and there exists } x \in \{-1, 0, 1\}^n \text{ with } Ax \leq b\}.$$

- a. Prove that $ILP \in NP$ by showing a polynomial-time verifier.
- b. Prove that $3\text{-SAT} \leq_p ILP$.